Candidate Number ________________ Fraction ________________

2022 33rd Republic of China Biology Olympiad National Team Selection Camp, Animal Anatomy, Systematics and Ecology, Third Examination

Room Practical **Exercise A**

All necessary equipment and materials for this examination have been placed on the table. Please carefully check all items listed on the checklist. If any are missing, Please raise your hand immediately to report to the reviewer.

I. Instructions for Experimental Apparatus:

Equipment 1 (Please leave it on the table when changing scenes)		Equipment 2 (One set per person, not to be reused)	
name	quantity	name	quantity
Pointed tweezers, pointed scissors, dissecting large paper cutters, rags, oil-based markers	2 pins (2 white, 2 black), 1 dissecting tray, 1 quail, 1 packet, 1 strip.	Distilled water (placed in a wash bottle)	4 sticks, 1 piece, 1 bottle

ÿImportant Notesÿ

1. The total time allotted for this exam, including both Practical Exercise A and Practical Exercise B, is 90 minutes. 2. Please ensure that your exam number on each page is correct; if there is an error, please raise your hand and ask the teaching assistant for assistance. 3. Materials, medicines, equipment, and anatomical samples on the table will not be replenished after use. 4. Please be mindful of safety. In case of an accident, you must notify the invigilator present for assistance. 5. This exam paper (including cover and question paper) consists of 3 pages, plus 2 pages of answer sheet. All pages must be returned upon submission. 6. Please answer on the answer sheet.

Experimental Topic: Observation and Comparison of the Urogenital System in Birds

The Japanese quail (Coturnix japonica) is a species of quail found in East Asia, classified as Aves, Coleoidea. (Galliformes), family Phasianidae, genus Coturnix, originally considered a subspecies of quail, was reclassified in 1983. Recognized as a separate species within the genus *Quercus* of the family Phasianidae. The Japanese quail is timid by nature and prefers to live in pairs in open, vegetated plains. Wilderness, pastures, and farmland environments. Small in size and with excellent camouflage, it often hides at the base of plants in farmland and grasslands. Japan Quails are easy to raise, grow quickly, and lay a large number of eggs. They are farmed all over the world, including in Taiwan, Japan, India, and China. Japanese quail are commercially farmed in the United States, Italy, Russia, and the United States. Japanese quail can provide a stable source of protein. The source is an ideal poultry alternative to chickens, not only for meat but, more importantly, the Japanese quail raised there can produce up to three hundred eggs per year. Each egg has considerable economic value.

The cloaca is a single posterior opening in birds that connects their digestive, urinary, and reproductive tracts. It is used for expelling feces and laying eggs. Located beneath the base of the tail, it is covered by feathers. Depending on the bird species, feces and urine may be stored in different chambers of the cloaca. In birds, solid material is typically stored in the innermost chamber as nutrient absorption continues. After digestion is complete, liquid and solid matter are separated. Waste is mixed together and discharged at the same time.

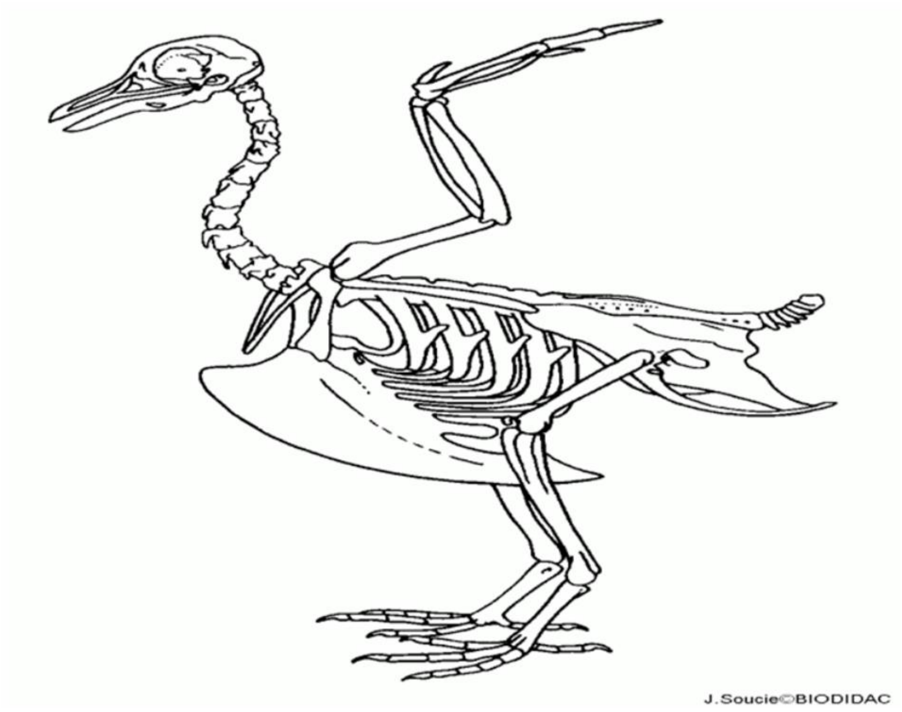
One quail is provided in the experimental materials. Please observe and compare the urinary and reproductive systems.

Precautions for experimental operation:

1. Please be careful when using dissection instruments. If injured, immediately raise your hand to inform the invigilator for assistance. 2. Before the exam ends, please place your answer sheet along with the organ structures described on it into the designated dissection tray.

The score will be awarded after the competition; otherwise, no score will be given.

Please remove the skin from the ventral side of the quail's body, but leave about 0.5 cm of skin around the cloaca.



Bird skeletal system diagram

1. Referring to the image above, remove the pectoral muscles from the bone to expose the keel, sternum, and ribs. The keel has fused with the sternum. Separate the joints of the sternal and spinal ribs. Separate the coracoid bone from the shoulder joint. Move the sternum, along with the sternal and coracoid bones, together.

   Remove the abdominal muscle wall without damaging the cloaca, exposing the abdominal viscera. Place the pectoral muscles and the sternum connected to the ribs and coracoid bone in the corresponding squares on Answer Sheet 1, and wait for the teaching assistant to grade them after the competition.

2. Completely remove the reproductive system and arrange it in the direction of egg production and expulsion (Aÿ Bÿcloaca). Place each organ on the answer sheet. Write the name in the corresponding square and wait for the teaching assistant to score it after the competition.

3. Locate at least three eggs at different stages of development within the reproductive system passage, and retrieve them from left to right according to their developmental stage. Arrange them in the corresponding squares on Answer Sheet 1, and wait for the teaching assistant to score them after the competition.

4. Locate the quail's urinary system. Mark the following structures on the quail with colored pins according to the direction of urine production and excretion. Wait for the teaching assistant to score the work after the competition, and write the names in the fill-in-the-blank section of the answer sheet.

5. Draw the complete urinary and reproductive system of a quail on Answer Sheet 2 and label each organ, as well as the orientation of all openings within the cloaca. Determine the relative position and label it with a name.

2022 Selection Camp Practical Third Test Student Code:

Candidate Number ______________________ Fraction ______________________

2022 33rd Republic of China Biology Olympiad National Team Selection Camp, Animal Anatomy, Systematics and Ecology, Third Examination

Session, Practical **Exercise B**

All necessary equipment and materials for this examination have been placed on the table. Please carefully check all items listed on the checklist. If any are missing, Please raise your hand immediately to report to the reviewer.

I. Experimental Equipment Description:

Equipment and Materials (Please leave them on the table when changing rooms)		Equipment and Materials (One set per person, no reuse)	
name	quantity	Quantity Name	
Dissecting microscope,	1 unit,	Dubia cockroach	2
dissecting tray (for insects), 10	1 piece,	detergent bottle (shared with Experiment A)	
insect needles, small	1 can,	Facial tissues (shared with Experiment A)	
dissecting	1 bunch,		
scissors,	1 bunch,		
forceps, beaker (waste	1 piece,		
cup), dropper	1 root		

ÿImportant Notesÿ 1. The total time allotted for this exam, including both Practical Exercise A and Practical Exercise B, is 90 minutes. 2. Please ensure that your exam number on each page is correct; if there is an error, please raise your hand and ask the teaching assistant for assistance. 3. Once the materials, medicines, equipment, and anatomical samples on the table are used up, they will not be replenished. 4. Please be mindful of safety; in case of an accident, you must notify the invigilator present for assistance. 5. This exam paper (including the cover and question paper) consists of 5 pages and must be returned in its entirety upon submission. 6. Please answer on the answer sheet.

Experiment Topic: External and Internal Structure of Cockroaches

The Dubia cockroach (Blaptica dubia) is native to Argentina. Females are wingless, while males are winged; they are ovoviviparous. The larvae hatch inside the female before being born. Due to their ease of care, Dubia cockroaches have long been common pets and are also an ideal food source for other animals. Two Dubia cockroaches are provided in this experiment; please perform dissection and observation of their external and internal morphology.

Precautions for experimental operation:

1. Please be careful when using the dissection instruments. If you are injured, immediately raise your hand to inform the invigilator for assistance. 2. After dissecting the head structure, please raise your hand immediately to request a score. Then, before the exam ends, please leave your answer sheet along with the cockroach on the dissection tray on the table for post-exam scoring; otherwise, no points will be awarded.

Operation process and issues:

1. Please use dissecting scissors or an insect needle to separate the head from the thorax, then dissect the cockroach's mouth parts. Remove the mandibles, labrum and clypeus, maxilla, and labrium from the skull. Refer to Figure 1 for each structure. Arrange the dissected structures on a dissecting tray as shown in Figure 2. Note that you must only use the insect needle to cut open the intersegmental membranes; do not directly cut off these appendages with dissecting scissors (20%). Raise your hand immediately after completing this part to request a grade.

2. After completing the above steps, please answer the following questions:

(1) How many segments are there on the maxillary palpus of the Dubia roach? Answer: ____________ (5%) (2) How many segments are there on the labial palpus of the Dubia roach? Answer: ____________ (5%) 3. Next, please wrap the mouthparts and head in tissue paper and discard them. Fix the cockroach's thorax to abdomen onto the dissection tray using an insect pin. Add water to the dissection tray using a washing bottle, ensuring the water completely submerges the cockroach's body. Cut open the dorsal skeletal plates, remove the muscles, trachea, fat body, and digestive tract, leaving the nerve cord intact, being careful not to break the body segments. You can use a washing bottle to clean the cockroach during the dissection. Raise your hand to request a score after completing this step. (15%) (Please refer to Figure 3)

4. How many thoracic ganglia does the Dubia roach have? _____________ (5%) 5. How many ventral ganglia does the Dubia roach have? _____________ (5%) 6. How do you distinguish muscles, trachea, Malpighian tubules, and fat bodies during dissection? (5%)

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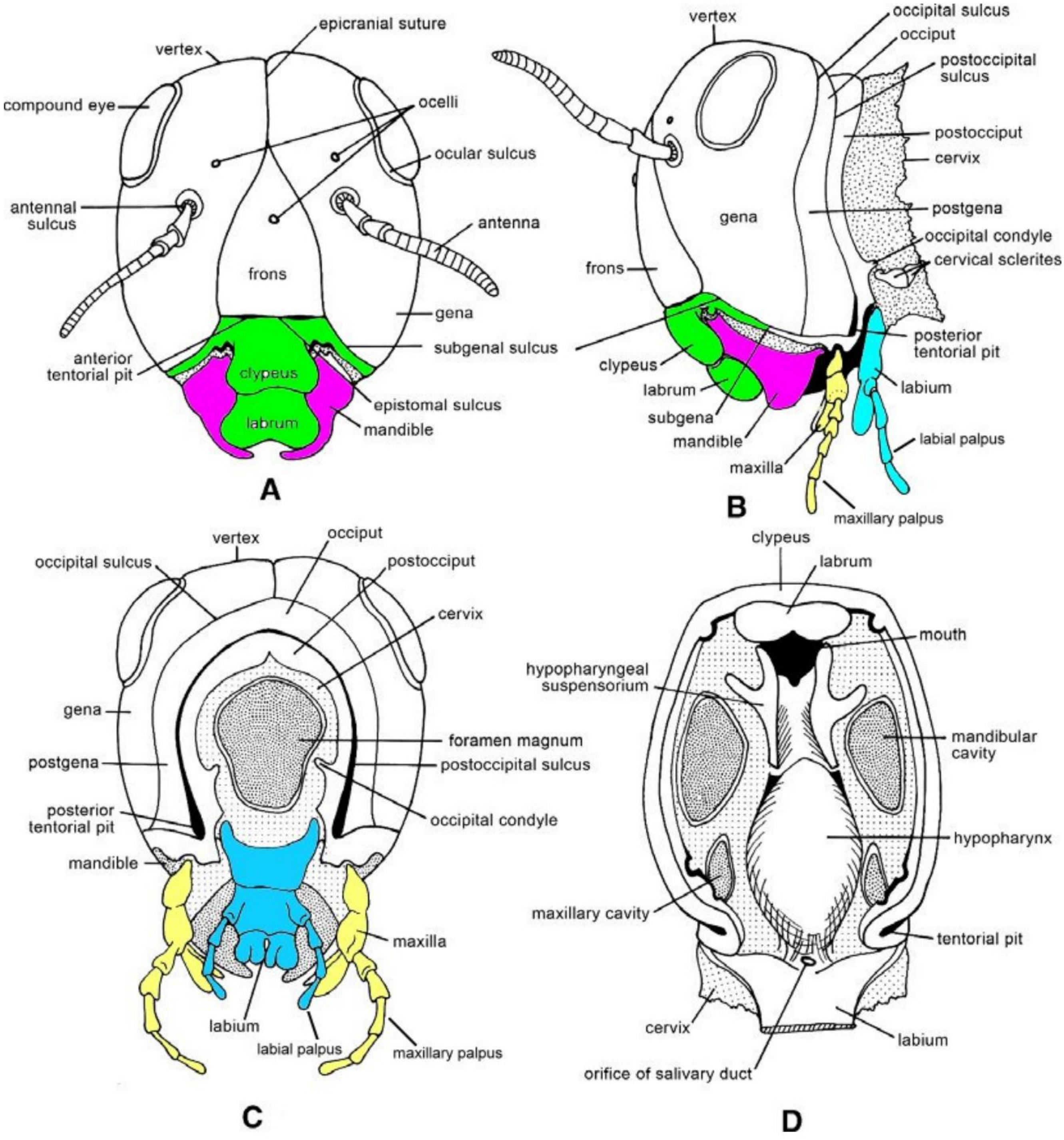

Figure 1. A structural diagram of the head of a chewing insect.

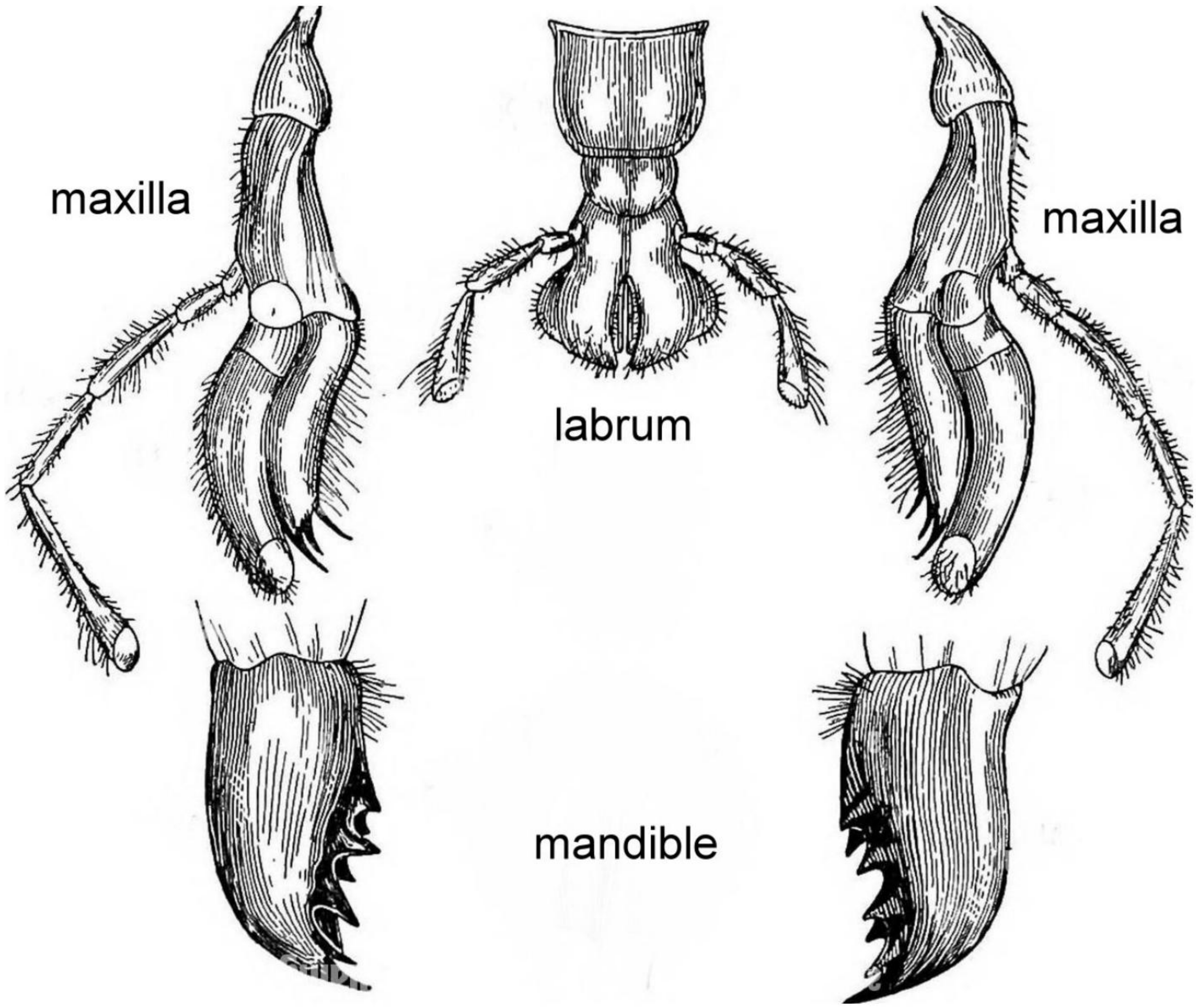

Figure 2. After dissecting the mouthparts, please arrange the parts in this manner, and then raise your hand to ask for a score.

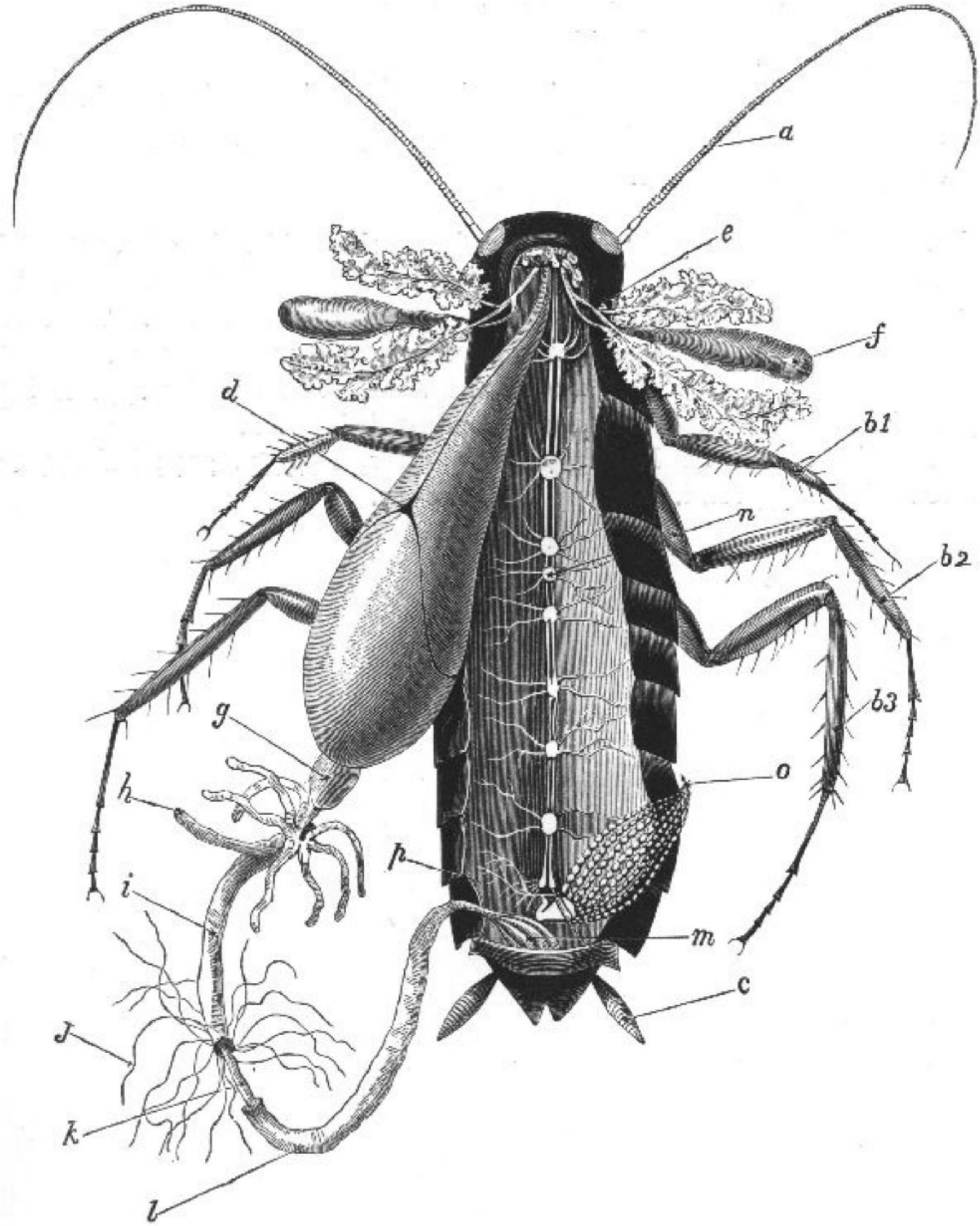

Figure 3. Schematic diagram of a cockroach body after removing muscle, trachea, fat body, and most Malpighian tubules. In Experiment B, please remove the entire digestive tract (d to l).